The largest Liouvillian exceptional point need not maximize the spectral gap: An exact two-qubit counterexample

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Liouvillian exceptional points can accelerate selected relaxation processes, but the exceptional point at the largest dissipation need not optimize the slowest asymptotic spectral rate. We establish this distinction exactly in a published two-qubit XYZ Lindblad model. At the fixed anisotropies $(\gamma, \Delta) = (3/5, 2/5)$, the largest positive Liouvillian exceptional point is $\Gamma_{\rm cr} = 8$. Exact Routh–Hurwitz calculations over the rationals nevertheless give

$$g(73/10) > \frac{3}{2} > 1 > g(8),$$

where g is the gap of the complete nonstationary Liouvillian spectrum. Thus $\Gamma_{\rm cr}$ is excluded from the set of global gap maximizers without a floating-point or branch-tracking assumption. This result concerns the state-independent asymptotic spectral rate and does not contradict exceptional-point optimization for specified quenches. The existence, location, and uniqueness of the actual global maximizer remain open.

I. INTRODUCTION

Exceptional points are non-Hermitian degeneracies at which eigenvalues and eigenvectors coalesce. In Markovian quantum dynamics, the full Liouvillian rather than a no-jump Hamiltonian determines such degeneracies [1]. When a slow decay mode participates in a Liouvillian exceptional point (LEP), tuning to the LEP can enlarge the Liouvillian gap and accelerate relaxation in experimentally relevant models [2]. The resulting optimization question is, however, objective dependent: the ordinary gap fixes an asymptotic spectral exponent, whereas eigenvector conditioning, Jordan prefactors, mode overlaps, and the chosen norm or initial state can control finite-time behavior [3].

Popkov, Presilla, and Salerno recently solved the spectrum of an open two-qubit XYZ model and described its LEP manifolds and effective Zeno regime [4]. For $(\gamma, \Delta) = (0.6, 0.4)$, their largest positive LEP is $\Gamma_{\rm cr} = 8$. Their relaxation comparison uses specified nonequilibrium-steady-state quenches and Hilbert–Schmidt distance. Here we ask the different, all-mode question of whether that largest LEP maximizes the complete nonzero spectral gap. A rational counterexample settles the exclusion part of this question: at $\Gamma = 73/10$, every nonstationary mode decays faster than $3/2$, while at $\Gamma = 8$ two modes decay more slowly than 1. The proof is a finite exact root count, not a numerical scan.

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II. MODEL, SYMMETRY, AND SCOPE

On $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$, consider the GKSL generator [5, 6]

$$\mathcal{L}_\Gamma(\rho) = -i[H, \rho] + \Gamma \left(L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\} \right), \tag{1}$$

$$H = \sigma^x \otimes \sigma^x + \frac{3}{5}\sigma^y \otimes \sigma^y + \frac{2}{5}\sigma^z \otimes \sigma^z, \tag{2}$$

$$L = \sigma^+ \otimes I_2. \tag{3}$$

Here $\sigma^+ = (\sigma^x + i\sigma^y)/2$, and I_2 is the one-qubit identity. We use the source normalization $J_x = \hbar = 1$ and restrict throughout to the physical half-axis $\Gamma > 0$. If the spectrum is counted with algebraic multiplicity, define

$$g(\Gamma) = -\max_{\lambda \in \mathrm{spec}(\mathcal{L}_\Gamma),\, \lambda \neq 0} \mathrm{Re}\, \lambda. \tag{4}$$

Let $S = \sigma^z \otimes \sigma^z$ and $\mathcal{P}(\rho) = S\rho S$. Although S anticommutes with L, conjugation leaves the dissipator in Eq. (1) invariant; also $[S, H] = 0$. Consequently $[\mathcal{P}, \mathcal{L}_\Gamma] = 0$, and Liouville space splits exactly into two eight-dimensional parity sectors, denoted Σ_+ and Σ_-. Direct construction from Eq. (1) agrees entry by entry with the vectorized source generator and reproduces its two characteristic factors.

The stationary eigenvalue is algebraically simple for every $\Gamma > 0$ in this fixed model. In the normalization used for the reconstructed factors, the derivative of the plus-sector characteristic polynomial at zero and the minus-sector value at zero are, respectively,

$$p'_+(0) = \frac{\Gamma^3(425\Gamma^2 + 1024)}{625} > 0, \tag{5}$$

$$p_-(0) = \frac{625\Gamma^6 + 19300\Gamma^4 + 203712\Gamma^2 + 112896}{15625} > 0. \tag{6}$$

Thus Eq. (4) unambiguously removes one zero root.

For reference, the discriminant of the source's plus-sector quartic reduces at the fixed anisotropies to

$$\frac{256}{390625}(\Gamma-8)^2(\Gamma+8)^2(5\Gamma-24)^2(5\Gamma+24)^2. \qquad (7)$$

The positive plus-sector branching values are therefore 24/5 and 8. Exact isolation of the minus-sector discriminant places all its positive branching values below 8. At $\Gamma = 8$, the plus-sector eigenvalues $-12/5$ and $-28/5$ each have algebraic multiplicity two, geometric multiplicity one, and squared-kernel dimension two. Hence 8 is a genuine LEP and reproduces the largest positive value $\Gamma_{\rm cr} = 8$ reported in Ref. [4].

III. EXACT RATIONAL EXCLUSION CERTIFICATE

For a real characteristic factor $p(\lambda)$ and a proposed decay threshold $r > 0$, set

$$q(s) = p(s-r), \qquad s = \lambda + r. \qquad (8)$$

All roots of p lie strictly to the left of $\mathrm{Re}\,\lambda = -r$ exactly when q is Hurwitz stable. For a real polynomial with positive leading coefficient, strict positivity of all leading principal Hurwitz determinants is necessary and sufficient for this property. More generally, when no first-column entry vanishes, the number of sign changes in the Routh first column equals the number of roots in $\mathrm{Re}\,s > 0$.

At $\Gamma = 73/10$, remove the unique stationary factor from p_+ and take $r = 3/2$. Clearing denominators and choosing positive leading coefficients gives the following exact coefficient vectors, ordered from the highest power of s to the constant term:

$$\begin{aligned}\mathbf{a}_{73,+} &= (40000000, 748000000, 5668600000,\\ &\quad 22630120000, 51613722250, 68431977325,\\ &\quad 51034832940, 17835041673),\\ \mathbf{a}_{73,-} &= (80000000, 1376000000, 9298000000,\\ &\quad 30896360000, 51691644500, 41256835900,\\ &\quad 18322078865, 8141339726, 592083829). \qquad (9)\end{aligned}$$

At $\Gamma = 8$ and $r = 1$, the required minus-sector vector is

$$\begin{aligned}\mathbf{a}_{8,-} &= (15625, 375000, 3647500,\\ &\quad 18405000, 51193350, 77020200,\\ &\quad 59217324, 28420344, 17748921). \qquad (10)\end{aligned}$$

These integer vectors are themselves compact certificates: the Hurwitz matrices and Routh tables follow by rational arithmetic alone. Their decisive sign data are shown in Table I. No zero Routh pivot occurs.

The two stable polynomials at 73/10 cover all fifteen nonstationary roots, so

$$\mathrm{Re}\,\lambda < -\frac{3}{2} \quad (\lambda \neq 0), \qquad g(73/10) > \frac{3}{2}. \qquad (11)$$

TABLE I. Exact sign certificates for the shifted sector polynomials. The last column is the number N_+ of roots in $\mathrm{Re}\,s > 0$. Repeated plus signs are denoted by $+^k$.

Γ	sector	Hurwitz determinant signs / Routh signs	N_+
73/10	+	$+^7$ / $+^8$	0
73/10	−	$+^8$ / $+^9$	0
8	−	$+^6, -, -$ / $+^7, -, +$	2

At $\Gamma = 8$, the two sign changes in the minus-sector Routh column show that exactly two roots obey $\mathrm{Re}(\lambda+1) > 0$. Therefore

$$g(8) < 1. \qquad (12)$$

Combining Eqs. (11) and (12) yields

$$\boxed{g(73/10) > \frac{3}{2} > 1 > g(8)}. \qquad (13)$$

The comparison point 73/10 belongs to the same physical domain. Equation (13) therefore excludes the reproduced largest LEP, $\Gamma_{\rm cr} = 8$, from the set of global gap maximizers.

IV. INDEPENDENT CHECKS AND PHYSICAL INTERPRETATION

Two independent generator constructions were used. The first transcribes the source's vectorized 16×16 matrix. The second applies Eq. (1) separately to all sixteen matrix units and never imports the first superoperator. They agree exactly after the explicit row/column vectorization permutation. The primary certificate then uses the parity blocks and Hurwitz determinants. A separate replay starts from the full matrix of the first construction, forms its degree-15 nonstationary polynomial, and builds a recursive Routh array without a parity split. Its first-column signs are $+^{16}$ at 73/10, and $+^{14}, -, +$ at 8, again giving zero and two right-half-plane shifted roots, respectively.

High-precision diagonalization provides only a diagnostic visualization: it gives gaps $1.5572286353\ldots$ at 73/10 and $0.9723556613\ldots$ at 8 (Fig. 1). These decimal values are not used in the proof; Eqs. (11) and (12) follow from integer signs alone.

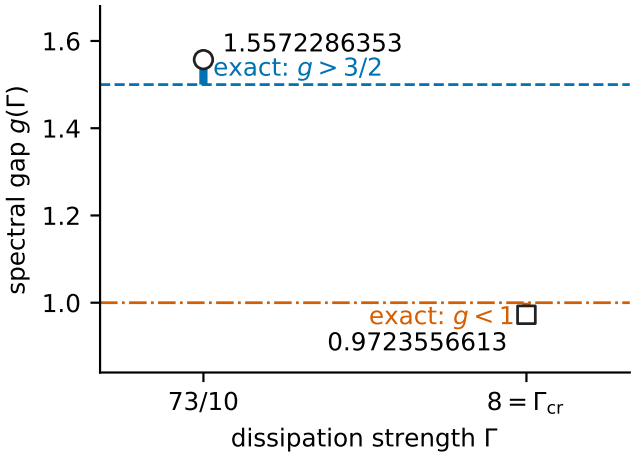


FIG. 1. Comparison at the two certified dissipation strengths. The circle and square are 105-digit diagnostic eigenspectrum values. The blue lower-bound segment and orange upper-bound segment encode $g(73/10) > 3/2$ and $g(8) < 1$, respectively; the numerical markers do not enter the certificate.

The gap in Eq. (4) is an all-mode asymptotic quantity. The exact comparison shows that tuning this model to its largest LEP sacrifices more than 1/2 in the guaranteed slowest exponential rate relative to the explicit nearby choice 73/10. It does not say that every state, observable, norm, or finite time window relaxes faster at 73/10. In particular, Jordan polynomials and mode overlaps at an LEP can favor a specified quench, which is the setting studied in Ref. [4]. The broader difference between ordinary gaps and finite-time relaxation is known [3]; the new content here is the exact fixed-model exclusion in Eq. (13).

V. CONCLUSION AND LIMITATIONS

An exact rational root count disproves largest-LEP optimality for the complete spectral gap of the fixed two-qubit generator. The result requires neither a floating-point optimizer nor a choice of active eigenvalue branch and applies to all fifteen nonstationary modes. Its scope is deliberately narrower than the original global optimization problem: we have not proved that the gap has a global maximizer on $(0, \infty)$, nor located or classified all maximizers. Establishing those facts requires a whole-axis certificate for branch changes and the small- and large-Γ tails.

DATA AVAILABILITY

The exact input matrices, integer-polynomial certificates, independent Routh replay, source code, runtime versions, and figure-generation script are provided as ancillary files with this manuscript. No experimental data were used.

ARTIFICIAL-INTELLIGENCE USE DISCLOSURE

Generative artificial-intelligence tools assisted with proposing the research question, literature screening, code generation, exact symbolic calculations, computational cross-checks, and manuscript drafting. They are not authors. The human author is responsible for independently reviewing, correcting, and approving all content before submission.

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